

PHYS 143 – Mechanics and Thermal Physics
Week 8: Rotation – Tutorial Sheet

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Use the Master Procedure: identify the body and the axis, sketch with the positive sense of rotation, choose the model (kinematics, $\sum \tau = I\alpha$, energy or angular momentum), predict, select the method, execute in radians, evaluate. Use $g = 9.80 \text{ m/s}^2$. Problems A1 and A2 are solved in full; final answers to the others are at the end.

A. Worked examples

A1. A grinding wheel

A grinding wheel of radius 0.12 m is accelerated uniformly from rest to 2400 rpm in 8.0 s and then runs at constant speed. Find (a) its angular acceleration, (b) the number of revolutions in the first 8.0 s, and (c) the speed and the radial acceleration of a point on the rim at $t = 8.0 \text{ s}$.

Solution. Anticlockwise positive. $\omega = 2400 \times \frac{2\pi}{60} = 251 \text{ rad/s}$ at $t = 8.0 \text{ s}$, $\omega_0 = 0$. (a) $\alpha = \omega/t = 251.3/8.0 = 31.4 \text{ rad/s}^2$. (b) $\theta = \frac{1}{2}(\omega_0 + \omega)t = \frac{1}{2}(251.3)(8.0) = 1005 \text{ rad} = 160 \text{ revolutions}$. (c) $v = r\omega = (0.12)(251.3) = 30.2 \text{ m/s}$; $a_{\text{rad}} = \omega^2 r = (251.3)^2(0.12) = 7.58 \times 10^3 \text{ m/s}^2$, about 773 g . Check: $a_{\text{rad}} = v^2/r = 30.2^2/0.12 = 7.6 \times 10^3 \text{ m/s}^2$. ✓

A2. Two blocks and a pulley with mass

Blocks of 5.0 kg and 3.0 kg hang from a light cord over a solid disc pulley of mass 4.0 kg and radius 0.15 m. The cord does not slip. Released from rest, find the acceleration of the blocks and the tension on each side.

Solution. Let the heavier block move down. $I = \frac{1}{2}MR^2$ and $a = R\alpha$. Blocks: $m_1g - T_1 = m_1a$ and $T_2 - m_2g = m_2a$. Pulley (the net torque in the sense of rotation): $(T_1 - T_2)R = I\alpha = \frac{1}{2}MR^2(a/R)$, so $T_1 - T_2 = \frac{1}{2}Ma$. Add the three equations: $(m_1 - m_2)g = (m_1 + m_2 + \frac{1}{2}M)a$, so $a = \frac{(2.0)(9.80)}{5.0 + 3.0 + 2.0} = 1.96 \text{ m/s}^2$. $T_1 = m_1(g - a) = 5.0(7.84) = 39.2 \text{ N}$; $T_2 = m_2(g + a) = 3.0(11.76) = 35.3 \text{ N}$. Check: $(T_1 - T_2)R = (3.92)(0.15) = 0.588 \text{ N m}$ and $I\alpha = \frac{1}{2}(4.0)(0.15)^2(13.07) = 0.588 \text{ N m}$. ✓ ($T_1 \neq T_2$ because the pulley has mass.)

B. Practice

B1. Angular position as a function of time

A rotor turns through $\theta(t) = 2.0t^2 - 0.10t^4$ (θ in radians, t in seconds). Find (a) ω and α at $t = 2.0 \text{ s}$, (b) the time at which the rotor momentarily stops (for $t > 0$) and the angle turned by then, and (c) its average angular velocity over that interval.

B2. A tyre

A car tyre of radius 0.32 m rolls without slipping at 90 km/h . (a) Find its angular velocity in rad/s and in rpm . (b) The car brakes uniformly to rest in 6.0 s with no slipping. Find the angular acceleration, the number of turns of the tyre, and the distance travelled.

B3. Four masses on a square

Four 0.50 kg masses sit at the corners of a light square frame of side 0.60 m . Find the moment of inertia about (a) an axis through the centre perpendicular to the frame, (b) an axis through one corner, perpendicular to the frame (check with the parallel-axis theorem), and (c) an axis along one side.

B4. Net torque

A rod can turn about a fixed axis O . Three forces act on it in the plane of rotation: 20 N perpendicular to the rod at 0.50 m from O (anticlockwise); 30 N at 0.80 m from O , at 30° to the rod (clockwise); and 15 N perpendicular to the rod at 0.20 m (clockwise). Find the net torque, and the angular acceleration if $I = 0.40\text{ kg m}^2$.

B5. A brake on a flywheel

A solid disc flywheel (40 kg , radius 0.50 m) turns at 600 rpm . A brake presses on the rim with a tangential friction force of 120 N . Find the time to stop, the number of revolutions, and the work done by the brake (check it against ΔK).

B6. Rolling down an incline

A thin hollow cylinder (2.0 kg , radius 0.10 m) rolls from rest without slipping 3.0 m down a 30° incline. Find its speed at the bottom (by energy) and its acceleration. Compare the speed with a frictionless sliding block.

B7. Walking to the centre of a turntable

A 60 kg person stands at the rim of a turntable (a uniform disc, 120 kg , radius 2.0 m) that turns at 0.80 rad/s on a frictionless axle. The person walks to the centre. Find the new angular velocity, the kinetic energy before and after, and the work done by the person. Treat the person as a point mass.

B8. Challenge: the method is not given

A uniform disc of mass 2.0 kg and radius 0.20 m spins at 30 rad/s about a vertical axle. A second, non-rotating disc with $I = 0.020\text{ kg m}^2$ is lowered on to it and sticks, so both turn together. The axle then exerts a constant frictional torque of 0.15 N m . Find the angular velocity just after the discs join, the kinetic energy lost, and the time and the number of turns for the pair to stop.

C. Concept questions

1. A wheel turns at constant ω . Is any point on it accelerating? Which point has the greatest acceleration, and in which direction?

2. Why do a solid sphere, a solid disc and a hoop of the same mass and radius reach the bottom of an incline in that order? Does the answer change if the mass is doubled?
3. The net force on a body is zero. Must the net torque be zero? Give an example.
4. A diver tucks into a ball during a somersault. What happens to her angular velocity, angular momentum and kinetic energy?
5. A rod hinged at one end is released from the horizontal. Is the tension in the hinge zero at the instant of release? Explain which forces act, and about which axis you take torques.

Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80 \text{ m/s}^2$.

Items E1 and E2 share this information: The angular position of a rotating wheel is $\theta(t) = 3.0t^2 - 0.40t^3$ (θ in rad, t in s).

E1. The angular velocity of the wheel at $t = 2.0 \text{ s}$ is

- (A) 1.2 rad/s
- (B) 6.0 rad/s
- (C) 7.2 rad/s
- (D) 8.8 rad/s
- (E) 12.0 rad/s

E2. Using your previous result, the speed of a point on the wheel 0.30 m from the axis at $t = 2.0 \text{ s}$ is

- (A) 0.36 m/s
- (B) 0.65 m/s
- (C) 2.16 m/s
- (D) 7.20 m/s
- (E) 24.00 m/s

Items E3 and E4 share this information: A grinder of constant moment of inertia 0.80 kg m^2 has angular velocity $\omega(t) = 2.0t^3$ (ω in rad/s, t in s).

E3. The net torque on the grinder at $t = 1.5 \text{ s}$ is

- (A) 3.6 N m
- (B) 5.4 N m
- (C) 7.2 N m
- (D) 10.8 N m
- (E) 13.5 N m

E4. The rotational kinetic energy of the grinder at $t = 1.5 \text{ s}$ is

- (A) 2.7 J (B) 3.6 J (C) 18.2 J (D) 36.5 J (E) 72.9 J

Items E5 and E6 share this information: A flywheel is a uniform solid iron disc (density 7800 kg/m^3) of thickness 10.0 cm , rotating about its central axis. It is to store $1.00 \times 10^6 \text{ J}$ of kinetic energy at an angular velocity of 25.0 rad/s .

E5. The diameter of the disc must be

- (A) 1.27 m (B) 1.80 m (C) 2.14 m (D) 2.54 m (E) 3.23 m

E6. The radial (centripetal) acceleration of a point on the rim at this angular velocity is

- (A) 31.8 m/s^2
(B) 397 m/s^2
(C) 625 m/s^2
(D) 795 m/s^2
(E) 1589 m/s^2

E7. A 0.40 kg particle is at $\vec{r} = (2.0\hat{i} - 1.0\hat{j} + 3.0\hat{k}) \text{ m}$ with velocity $\vec{v} = (4.0\hat{i} - 2.0\hat{j} + 5.0\hat{k}) \text{ m/s}$. Its angular momentum about the origin $\vec{L} = \vec{r} \times \vec{p}$ has magnitude

- (A) $0.40 \text{ kg m}^2/\text{s}$
(B) $0.89 \text{ kg m}^2/\text{s}$
(C) $1.20 \text{ kg m}^2/\text{s}$
(D) $5.02 \text{ kg m}^2/\text{s}$
(E) $10.04 \text{ kg m}^2/\text{s}$

E8. A 4.0 kg particle at $\vec{r} = (0.80\hat{i} + 0.60\hat{j}) \text{ m}$ (the xy -plane is vertical, $+y$ up) is acted on only by its weight. The rate of change of its angular momentum about the origin, $d\vec{L}/dt = \vec{r} \times \vec{F}$, is

- (A) $-31.4 \hat{k} \text{ N m}$
(B) $+31.4 \hat{k} \text{ N m}$
(C) $-23.5 \hat{k} \text{ N m}$
(D) $-31.4 \hat{j} \text{ N m}$
(E) 0

E9. Which of the following statements is/are true?

- I. The torque of a force about a point depends on the magnitude of the force and on the perpendicular distance from the point to the line of action of the force.
- II. The angular momentum of a system is conserved if the net external torque on it is zero.
- III. The moment of inertia of a rigid body does not depend on the position of the rotation axis.

- (A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

E10. A skater spinning at 2.0 rad/s with moment of inertia 3.0 kg m^2 pulls in her arms, reducing it to 1.2 kg m^2 . The ratio of her final to her initial rotational kinetic energy, K_f/K_i , is

- (A) 0.40 (B) 1.00 (C) 5.00 (D) 6.25 (E) 2.50

Answers

- B1 (a) $\omega = 4.8 \text{ rad/s}$, $\alpha = -0.80 \text{ rad/s}^2$; (b) $t = \sqrt{10} = 3.16 \text{ s}$, $\theta = 10.0 \text{ rad}$; (c) 3.16 rad/s
 B2 (a) 78.1 rad/s (746 rpm); (b) $\alpha = -13.0 \text{ rad/s}^2$; 37.3 turns; 75 m (equals $\frac{1}{2}(25)(6.0)$)
 B3 (a) 0.36 kg m^2 ; (b) $0.72 \text{ kg m}^2 (= 0.36 + 2.0(0.424)^2)$; (c) 0.36 kg m^2
 B4 $\sum \tau = +10.0 - 12.0 - 3.0 = -5.0 \text{ N m}$ (clockwise); $\alpha = -12.5 \text{ rad/s}^2$
 B5 $I = 5.0 \text{ kg m}^2$, $\omega_0 = 62.8 \text{ rad/s}$, $\tau = 60 \text{ N m}$, $\alpha = -12.0 \text{ rad/s}^2$; $t = 5.24 \text{ s}$; 26.2 turns (164 rad); $W = -9.87 \times 10^3 \text{ J} = \Delta K$
 B6 $v = \sqrt{gh} = 3.83 \text{ m/s}$ ($h = 1.5 \text{ m}$); $a = \frac{1}{2}g \sin 30^\circ = 2.45 \text{ m/s}^2$ (check $v^2 = 2as$); a sliding block would reach 5.42 m/s
 B7 $I_i = 480$, $I_f = 240 \text{ kg m}^2$; $\omega_f = 1.60 \text{ rad/s}$; $K_i = 154 \text{ J}$, $K_f = 307 \text{ J}$; work done by the person = 154 J
 B8 $\omega = 20.0 \text{ rad/s}$ ($L = 1.2 \text{ kg m}^2/\text{s}$ conserved); K : $18.0 \rightarrow 12.0 \text{ J}$, lost 6.0 J ; $\alpha = 2.5 \text{ rad/s}^2$, $t = 8.0 \text{ s}$, $80 \text{ rad} = 12.7 \text{ turns}$
 C1 yes: every point not on the axis has $a_{\text{rad}} = \omega^2 r$ toward the axis; the greatest at the rim
 C2 sphere ($c = \frac{2}{5}$), disc ($c = \frac{1}{2}$), hoop ($c = 1$); $a = g \sin \theta / (1 + c)$ does not contain M or R
 C3 no: a couple (two equal and opposite forces with different lines of action) gives zero net force and a non-zero torque
 C4 I decreases, so ω increases; L is unchanged; $K = L^2/2I$ increases (her muscles do work)
 C5 about the hinge, the hinge force has no torque, so $\alpha = \frac{3}{2}g/L$; the hinge force is not zero: $F_y = Mg - Ma_{\text{cm}} = Mg - M\alpha L/2 = \frac{1}{4}Mg$ upward at that instant

Exam-style practice: answer key and brief solutions

Key: E1 C E2 C E3 D E4 C E5 D E6 D E7 B E8 A E9 B E10 E

E1 (C). $\omega = d\theta/dt = 6.0t - 1.2t^2 = 12.0 - 4.8 = 7.2 \text{ rad/s}$. The value $\theta(2.0) = 8.8 \text{ rad}$ is the angle, and $\alpha = 6.0 - 2.4t = 1.2 \text{ rad/s}^2$ is the angular acceleration.

E2 (C). $v = \omega r = (7.2)(0.30) = 2.16 \text{ m/s}$, tangent to the circle. Check units: $\text{rad/s} \times \text{m}$ gives m/s (the radian is dimensionless). The tangential acceleration is $\alpha r = 0.36 \text{ m/s}^2$.

E3 (D). $\alpha = d\omega/dt = 6.0t^2 = 13.5 \text{ rad/s}^2$ at $t = 1.5 \text{ s}$, and $\sum \tau = I\alpha = (0.80)(13.5) = 10.8 \text{ N m}$. The value $I\omega = 5.4$ is an angular momentum, and $\alpha = 13.5$ is not yet a torque.

E4 (C). $\omega(1.5) = 2.0(1.5)^3 = 6.75 \text{ rad/s}$, so $K = \frac{1}{2}I\omega^2 = \frac{1}{2}(0.80)(6.75)^2 = 18.2 \text{ J}$. Check by work: K is the work done by the torque, $\int \tau d\theta = \int I\alpha \omega dt$.

E5 (D). Chain: $K = \frac{1}{2}I\omega^2$, $I = \frac{1}{2}MR^2$, $M = \rho\pi R^2 t$, so $K = \frac{1}{4}\rho\pi t\omega^2 R^4$. Then $R^4 = \frac{4K}{\rho\pi t\omega^2} = 2.612 \text{ m}^4$, $R = 1.271 \text{ m}$ and $D = 2.54 \text{ m}$. Check: mass $M = 3960 \text{ kg}$, $I = 3200 \text{ kg m}^2$, $\frac{1}{2}I\omega^2 = 1.00 \times 10^6 \text{ J}$. The value $R = 1.271 \text{ m}$ is the radius.

E6 (D). $a_{\text{rad}} = \omega^2 R = (25.0)^2(1.271) = 795 \text{ m/s}^2$, about 81 times g : the rim is highly stressed. The value $\omega R = 31.8 \text{ m/s}$ is the rim speed, not an acceleration.

E7 (B). $\vec{p} = m\vec{v} = (1.6, -0.8, 2.0)$, and $\vec{L} = \vec{r} \times \vec{p} = [(-1)(2.0) - (3)(-0.8)]\hat{i} - [(2)(2.0) -$

$(3)(1.6)]\hat{j} + [(2)(-0.8) - (-1)(1.6)]\hat{k} = (0.4\hat{i} + 0.8\hat{j} + 0\hat{k}) \text{ kg m}^2/\text{s}$. $|\vec{L}| = 0.89$. Check: $\vec{L} \cdot \vec{r} = 0.8 - 0.8 = 0$ and $\vec{L} \cdot \vec{p} = 0.64 - 0.64 = 0$, as required for a cross product. The product $rp = 10.04$ is the maximum possible $|\vec{L}|$ (for $\vec{r} \perp \vec{p}$).

E8 (A). $\vec{F} = -mg\hat{j} = -39.2\hat{j} \text{ N}$ and $\vec{r} \times \vec{F} = (0.80)(-39.2)\hat{k} - (0.60)(0)\hat{k} = -31.4\hat{k} \text{ N m}$ (magnitude $x F_y$ only, the lever arm is the horizontal distance 0.80 m). The $-\hat{k}$ direction is into the page for x right, y up: a clockwise torque, as sketching the weight acting to the right of the origin shows. The $\vec{r} \times \vec{F}$ vector is perpendicular to the xy -plane, so it cannot point along $\hat{j}$.

E9 (B). I is true: $\tau = rF \sin \phi = F\ell$ with ℓ the lever arm. II is true: $d\vec{L}/dt = \sum \vec{\tau}_{\text{ext}}$. III is false: $I = \sum m_i r_i^2$ depends on the axis (parallel-axis theorem $I = I_{\text{cm}} + Md^2$). Answer: I and II only.

E10 (E). No external torque, so $I_1\omega_1 = I_2\omega_2$ and $\omega_2 = 5.0 \text{ rad/s}$. Then $K_f/K_i = \frac{I_2\omega_2^2}{I_1\omega_1^2} = \frac{(1.2)(25)}{(3.0)(4.0)} = 2.5$; equivalently $K = L^2/2I$ rises because I falls. The extra energy comes from the work done by her muscles pulling the arms inward. The ratio 5.0 is ω_2/ω_1 .